The Youth Science Journal Of Iraq

Issue #8 May 2026

First Photo Feature

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The Challenge Solution:

Last month's April Fool's Edition featured a false article; the very last one, that is Question VI: Why is the Moon's density so low?

Luckily for us, the moon is not in fact made of aerogel nor does it float on solar winds. It is instead made of low density silicate rock, stemming from its origin.

A Short Editorial:

YSJOI - by the youth, for the youth of Iraq!

In the great land that was once Mesopotamia, and the home of the hanging gardens of Babylon, progress in science has slowed.

Why?

Because the youth of Iraq have no platform to build upon! Opportunities to learn about science, technology, engineering, and mathematics are rare, and students trying to learn about them are even actively discouraged – I speak from experience.

But instead of surrendering to that silence, I have chosen to rally the youth of Iraq. If we do not have a base to stand on, we will build one!

And the first step to that is this Youth Science Journal Of Iraq, YSJOI, where we aim to provide that very base – an opportunity to teach, learn and explore through different articles – be it inspiring them, writing them, or reading them!

Thank you for being part of this beginning.
Now, dive in – into the weird, the beautiful, and the wonderful.

– Satvik M. Bhure, Founder of YSJOI

Question I: When did the first humans evolve and how?

– Ahmed Ali, 15

The first humans evolved in Africa through a slow, messy process of trial and error that unfolded over millions of years. Around six to seven million years ago, the lineage that would become humans split from the one that led to chimpanzees and bonobos. From that branching point, a succession of different species experimented with new body shapes, diets, and behaviors; some walked more upright, some developed different teeth and jaws, and some learned to use simple tools. Over time, the successful combinations accumulated into the traits we now recognize as human.

One of the earliest and most important changes was bipedalism. Walking on two legs altered the pelvis and spine, changed how energy was used during movement, and freed the hands for carrying, holding infants, and manipulating objects. That freedom made new behaviors possible and opened ecological opportunities across Africa's shifting landscapes. As posture and locomotion changed, so did diet and technology: stone flakes and cut marks on bones show that hominins were making and using tools as far back as about 3.3 million years ago, long before people looked like us. Tool use and food processing fed back into evolution, favoring hands capable of fine manipulation and brains able to learn from others.

By roughly 2.8 million years ago, fossils assigned to the genus Homo appear, showing a trend toward larger brains, smaller teeth, and longer childhoods—conditions that support learning, teaching, and cultural accumulation. Over hundreds of thousands of years, brain size increased, toolkits became more varied, and social groups grew more complex. These biological and cultural changes together produced people who looked and behaved more like us.

Anatomically modern Homo sapiens appeared in the African fossil record around 300,000 years ago. Becoming human was not a single moment but a mosaic process: different hominin groups coexisted, migrated, and sometimes interbred. Neanderthals and Denisovans, for example, left genetic traces in many modern populations, showing our ancestry is braided rather than linear.

New fossils, tools, and ancient DNA keep refining this story. What remains clear is that humans emerged gradually through many small changes driven by environment, technology, and social life—not by one dramatic leap.

Question II: What would the universe be in reverse?

– Novar Mazhar, 12

We have all heard that electrons are negatively charged and that protons are positively charged. You might have also heard that non-metal ions are usually negatively charged and that metal ions are positively charged. Why is this?

Let's take chlorine as an example. A chlorine atom has 17 protons and 17 electrons, so overall it is neutral. However, the atom is very unstable due to its electronic configuration (2,8,7). For particles to be stable, they need to have an electronic configuration of that of a noble gas.

The nearest noble gas to chlorine is argon, which has the electronic configuration (2,8,8). Therefore, chlorine tries to get one electron to be stable. It can do this in many ways, one of which is becoming a chlorine ion, but where does it get this electron?

A metal such as sodium has the electronic configuration (2,8,1), so it wants to lose an electron to be stable, meaning it will have the electronic configuration of neon, the nearest noble gas to it. Thus, if we react sodium with chlorine, the electron from sodium will transfer to the chlorine, producing sodium chloride, which consists of two ions, those being a positive sodium ion and a negative chloride ion.

However, this raises the question of why an electron is negative to begin with.

There is, in fact, nothing that makes an electron negative; it is just a naming convention, as when Benjamin Franklin discovered them, he assigned each their respective charges. So what would happen if we decided to make an electron positive and a proton negative? At first glance, it seems to make chemistry and physics easier to understand. Mainly, conventional current would be identical to electron flow (if electrons were positively charged).

However, to switch the signs of protons and electrons would be rewriting the system many were taught and raised by and would overall matter little for such a grand project. In the end, such things are only labelled 'positive' or 'negative' to aid in our understanding of the world.

Question III: Could we move to other planets?

– Dila Mustafa, 17

Since the early beginning of human history, colonization has been a common, although unethical, practice. Countries find land and claim it as their own, drawing up borders to preserve it. We've mapped out most of the surface of the Earth as of now. Why not extend our reach to outer space? A common movie theme is space survival: astronauts leaving Earth on a mission to Mars, living in a spacecraft and in their spacesuits almost all the time. What if we could live there without the suits? Let's assume we had the means to travel to space, what would set us back?

One major setback is the atmosphere. Our atmosphere contains the right amount of required gases (~78% nitrogen, ~21% oxygen, and ~1% others). The atmospheres of other planets have no such mercy on us. Venus, for example, contains an atmosphere ruled by carbon dioxide, which makes up 96.5% of it. Nitrogen makes up the rest, along with trace amounts of other gases, and a very small concentration of breathable oxygen. Even if we were to reach it, we wouldn't be able to make use of the variation of gases nor breathe without spacesuits, hence why living there is a deadly challenge.

Another wonderful trait of Earth is the magnetic field surrounding it; caused by the movement of magnetic liquids in its core. This is called a core dynamo. Meanwhile, the Red Planet, also known as Mars, lost its core dynamo almost 4 billion years ago. Consequently, it now has a very weak magnetic field, making it vulnerable to solar winds and cosmic radiation. That destroyed Mars' atmosphere and made it uninhabitable for us. If we were exposed to that kind of radiation, our cells could mutate and possibly lead to various problems. We simply wouldn't be able to live safely.

The final characteristic of Earth is its place in the solar system. It's perfectly positioned for liquid water (vital for survival) to exist naturally. Other planets are either too far or too close to the Sun. The temperature difference would be highly inconvenient, if not fatal. If we were to travel there, we would die.

To conclude, Earth's composition, complexion, and gifts were perfectly made to accommodate life as we define it. Moving to other planets would be deadly for us— considering our human requirements and limits.

Question IV: How could we time travel?

– *Sitalaxmi Narayan, 9*

Time travel is a very wide and fascinating concept, one discussed and shown in many ways over time, an idea that has intrigued us for countless years.

Into the future, there are two main ways that have been proven in real-life experiments to work successfully, both relying on the same core concept of Einstein's relativity. To better understand this, think of the fabric of reality as a bucket that holds variable amounts of everything it holds, the main things being space and time, another very important factor to consider: the absolute maximum speed possible is the speed of light, light will constantly move away from us at the same speed, it is also impossible to reach the exact speed of light, the speed of light is basically the maximum amount the bucket mentioned earlier can hold. The first way to travel ahead in time is by travelling so fast (about 99.9% of the speed of light) that we nearly fill up the bucket to a point where the amount of time inside is forced to decrease, therefore slowing down time, gravity stays constant. The second way is by increasing gravity to an extreme point so that this theoretical bucket is also filled so that time has to slow down to compensate once again, this can be done by orbiting the boundary of a black hole. Time appears to pass normally to any person doing this, but travelling one hour at the speed of light might be a few years gone on Earth. Noting that traveling forward into time is a one-way ticket and there are no paths to return currently.

Into the past, this is a lot more complex due to the paradoxes that could be caused. Most believe backwards time travel is impossible; however, theories do exist to make this phenomenon possible. The most major is the concept of wormholes, a wormhole is like a tunnel of space looped around itself to have two openings, one in the present and one that leads to another location. If one gate were to moving at near the speed of light and the other stationary, by crossing through the moving one a person could arrive at the other gate in the past, but this requires negative energy which we have not yet found a way to find or control.

Time traveling is a thing of sci-fi movies, but physicists have proved that it is a credible idea, at least in part.

Question V: What is one big advancement in human history?

– *Daryan Muhammed, 13*

One of the biggest advancements in human history has to be the invention of the internet. Some people may argue that electricity, medicine, or even the Industrial Revolution were more important. And rightfully so, all of the above have helped shape events beyond the human brain. However, the internet in particular stands out because of its influence on almost every part of modern life within a short period of time.

It is hard to imagine a world without the internet, its counterparts, and their impact on a day-to-day basis. For example, communication: It used to take a long time to get a message across to another person; people relied on letters, newspapers and limited forms of media to stay informed. Now, news spread globally in a matter of seconds, and people can connect instantly no matter where they are. Families separated by distance can still stay close through video calls and instant messages, which is great evidence in favor of how much human communication has advanced with the help of the internet.

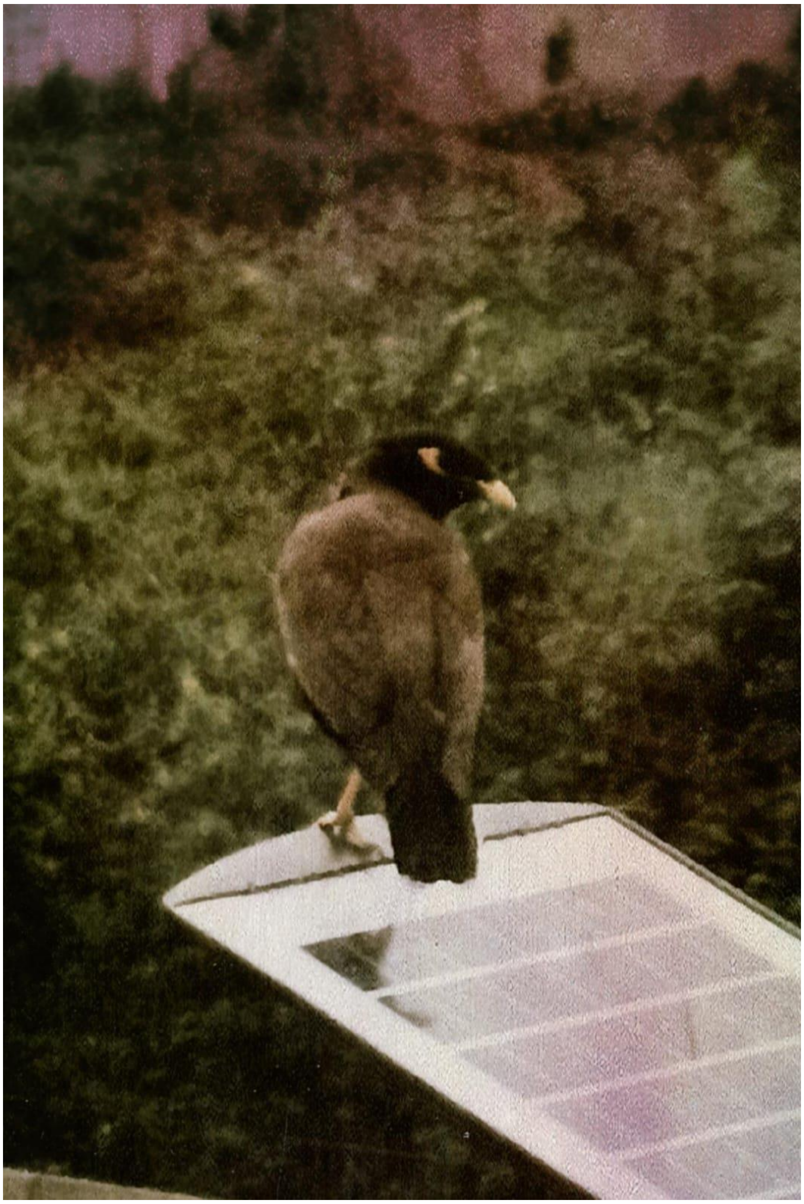
Another major impact of the internet is education. In the past, learning opportunities were often limited by location or money. Today, students can access lectures, articles and courses from almost anywhere. Any person with a digital device and an internet network can learn skills that once required money in return or were only available in libraries. Therefore, the internet has made knowledge far more accessible than before.

The internet has also transformed businesses and daily life. Shopping, banking and even working can now be done online. Many people earn payment through digital platforms that did not exist twenty years ago. At the same time, companies can reach millions of people globally instead of only focusing on local customers.

It is only fair to state that the internet is not perfect. Conflicts arise; misinformation, online addiction and privacy issues have become serious concerns. Even so, its overall influence on humanity is impossible to ignore.

First Photo Feature

This magnificent image of the Indian Myna (Acridotheres tristis) was captured by one of our staff. We urge you to send your best pictures to us as well, so we may feature it as well!



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For giving us the chance to answer their questions in the May 2026 Edition of YSJOI!

Thanks also to the growing team of YSJOI, where we have welcomed many new members, all of who had a hand in this incredible issue and all the work that went into it:

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Asal Ibrahim

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Abdullah Haider

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Hasar Salah

And we thank you, reader, for supporting our mission to help the Iraqi youth where there has been no help as yet!

CONCLUSION

We hope to further YSJOI's mission to empower the youth of Iraq in STEM fields by our monthly magazine, and sincerely hope we were able to help you and other students with this monthly issue! We are always open to feedback and questions, and everyone is of course welcome to learn in their own way; after all, we are all different, and we all belong.

Make sure to visit our website at https://YSJOI.is-great.org

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Or DM us at www.instagram.com/ysjoi_official